ELECTRONIC SUPPLEMENTARY INFORMATION (ESI)

Full MHD Derivation Proofs and Buckingham Pi Scale-Invariance Metrics

Prepared for *Fusion Engineering and Design*

Jacob James Rasmussen[1,2]

[1]Independent Researcher, Shell Lake, WI 54871, USA

[2]Incoming STEM Transfer Program Scholar, Chippewa Valley Technical College (CVTC), Eau Claire / Shell Lake, WI

ORCID: 0009-0009-7084-6523

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1 3D Navier-Stokes-Maxwell MHD Boundary Layer Proofs

Under steady, incompressible flow in a rectangular duct with transverse magnetic field $\mathbf{B} = B_0\hat{\mathbf{z}}$, the momentum equation reduces to:

$$0 = -\frac{dp}{dz} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + J_y B_0 \tag{1}$$

Using Ohm's Law $J_y = \sigma(-\frac{\partial \phi}{\partial y} - uB_0)$, the Hartmann velocity distribution across normalized width y/a is derived as:

$$u(y) = u_{\text{core}}\left(1 - \frac{\cosh(Ha \cdot y/a)}{\cosh(Ha)}\right) \tag{2}$$

2 Buckingham Pi Scale-Invariance Analysis

System dimensional parameters $(\rho, \mu, \sigma, u_0, B_0, a, c_w)$ form four independent dimensionless groups according to Buckingham's Pi theorem:

Table 1: Buckingham Pi Dimensionless Group Metrics for Liquid Metal Divertors

Dimensionless Group	**Formula**	**Benchmark Value (Pb-17Li)**
Π_1 (Hartmann Number)	$Ha = B_0 a\sqrt{\sigma/\mu}$	500.0
Π_2 (Interaction Parameter)	$N = \sigma B_0^2 a/(\rho u_0)$	1250.0
Π_3 (Reynolds Number)	$Re = \rho u_0 a/\mu$	9.3×10^5
Π_4 (Wall Conductance Ratio)	$c_w = \sigma_w t_w/(\sigma a)$	0.01

3 MHD Pressure Drop Scaling

For electrically conducting walls with conductance ratio c_w, the dimensional pressure gradient dP/dz scales as:

$$\frac{dP}{dz} = -\sigma u_0 B_0^2 \left(\frac{c_w}{1 + c_w} \right) = -1.0 \times 10^6 \times 1.0 \times 100 \times \left(\frac{0.01}{1.01} \right) \approx 0.45 \text{ MPa/m} \tag{3}$$